

TSEGAY TESFAY MEZGEBE, PhD

Associate Professor of Industrial & Systems Engineering (Automatic Control of Manufacturing Systems)

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PROFESSIONAL SUMMARY

Associate Professor and engineering leader specializing in the design of automatically controlled manufacturing systems aligned with the Factory of the Future. Bridges scientific and industrial challenges across industrial and systems engineering, intelligent manufacturing, optimization, automatic control, and lean manufacturing, with proven experience in technology transfer and reverse engineering. Track record of leading internationally funded projects, delivering industrial consultancy, supervising doctoral and masters research, and publishing in leading peer-reviewed journals.

RESEARCH INTERESTS

Building automatically controlled manufacturing systems aligned with the requirements of the Factory of the Future, addressing scientific and industrial challenges in industrial and systems engineering, intelligent manufacturing, optimization, automatic control, and lean manufacturing. Technology transfer and reverse engineering are complementary themes of interest.

Keywords: *Industrial and Systems Engineering, Automatic Control, Intelligent Manufacturing Systems, Optimization, Lean Manufacturing, Multi-Agent Systems, Modeling of Complex Industrial Processes, Technology Transfer, Industry 4.0*

EDUCATION

PhD in Automatic Control, CRAN, University of Lorraine, France (awarded January 15, 2020) 2017 – 2020

MSc in Production and Industrial Systems Engineering, Ethiopian Institute of Technology-Mekelle, Mekelle University, Ethiopia (awarded October 27, 2011) 2009 – 2011

BSc in Industrial Engineering, Faculty of Engineering, Bahir Dar University, Ethiopia (awarded June 06, 2007) 2003 – 2007

CERTIFICATIONS & PROFESSIONAL TRAINING

Seminar on “Poverty Reduction and Alleviation through Business and Trade for Developing Countries” – Ministry of Commerce of China & China Center for Urban Development (NDRC), Beijing, China May–Jun 2026

Master Class for Teaching Online (Open edX) – EdPlus Innovation and Emerging Initiatives, Arizona State University, USA Jan–Feb 2026

Artificial Intelligence Fundamentals – Udacity, USA Oct–Nov 2024

Data Analysis Fundamentals – Udacity, USA Sep 2024

Meta-Heuristics in Optimization: Current Trends and Evolutions – Ecole MACS 2019, University of Bordeaux, France Jun 2019

Cyber-Physical Production Systems – IUT de Nantes, University of Nantes, France Jan 2019

Culture of Scientific Integrity – University of Lorraine, France Mar 2018

Ethical and Legislative Framework for Research – University of Lorraine, France May 2017

Quality Management System Development & Implementation (ISO 9001:2008) – Ethiopian Standards Agency Mar 2015

Quality Management Systems Audit (ISO 19011:2011) – Ethiopian Standards Agency Apr 2015

Life Skills Training – Bahir Dar University, Anti-HIV/AIDS Movement Club, Ethiopia Apr 2007

ACADEMIC EXPERIENCE

Associate Professor, EiT-M, Mekelle University, Ethiopia	<i>Jan 2025 – Present</i>
Honorary Adjunct Professor, Chandigarh Group of Colleges, Jhanjeri, Mohali, India	<i>Jun 2025 – Present</i>
Assistant Professor, EiT-M, Mekelle University, Ethiopia	<i>Mar 2020 – Jan 2025</i>
Postdoctoral Researcher, CRAN, University of Lorraine, France	<i>Dec 2019 – Mar 2020</i>
Assistant Professor, EiT-M, Mekelle University, Ethiopia	<i>May 2016 – Jan 2017</i>
Lecturer, EiT-M, Mekelle University, Ethiopia	<i>Nov 2011 – Apr 2016</i>
Assistant Lecturer, EiT-M, Mekelle University, Ethiopia	<i>Aug 2008 – Sep 2011</i>

Key responsibilities and contributions:

- Supervisor for 3 doctoral students (automatic control and lean manufacturing) and co-supervisor for 1 doctoral student (quality infrastructure 4.0) at Mekelle University.
- Supervised 9+ masters students in optimization of complex industrial systems, industrial automation and mechatronics, manufacturing and composites, and lean manufacturing.
- Supervised 10+ bachelor students in industrial, manufacturing, and mechanical engineering.
- Conducted research on distributed manufacturing control using multi-agent systems during postdoctoral tenure at CRAN, collaborating with researchers at the University of Lorraine, France.
- Taught doctoral courses in industrial automation, CIM, lean manufacturing, and advanced research methodologies.
- Taught masters courses in automation systems and manufacturing control, advanced manufacturing processes, and industrial and systems engineering.
- Taught bachelor courses in industrial, mechanical, and manufacturing engineering.
- Strong engagement in internationally and locally funded projects and research.

INDUSTRY EXPERIENCE

Director, Beles Tech Research and Development, Addis Ababa, Ethiopia – administering R&D on prickly pear cactus
Nov 2024 – Jun 2026

Acting Head, Merchandizing Division, MAA Garment and Textile Factory, Mekelle, Ethiopia – managing the merchandizing division under the marketing department
Jun–Aug 2008

Merchandizer, MAA Garment and Textile Factory, Mekelle, Ethiopia – following all production activities in the garment plant
Mar–Jun 2008

ACADEMIC LEADERSHIP

Director, International Affairs Directorate, Mekelle University, Ethiopia
Jun 2026 – Present

Dean, School of Mechanical and Industrial Engineering, EiT-M, Mekelle University, Ethiopia
Jun 2021 – Oct 2025

Head of Postgraduate Studies, EiT-M, Mekelle University, Ethiopia
Apr 2020 – Jun 2021

Head of Quality Assurance Office, EiT-M, Mekelle University, Ethiopia
Dec 2014 – Jan 2017

Head of Manufacturing Engineering Chair, School of Mechanical and Industrial Engineering, EiT-M, Mekelle University, Ethiopia
May 2013 – Nov 2014

FUNDED PROJECTS

Local Coordinator, “Strengthening Mobility and Promoting Regional Integration of Engineering Education in Africa (SPREE)” – €1,392,975 funded by the European Commission (spreeafrica.com)
2023 – 2025

Research Assistant, “The Contribution of Vocational Skills Development to Inclusive Industrial Growth and Transformation, Phase 2” – CHF 2.8 million funded by SDC and the Swiss National Science Foundation (SNSF)
2021

Project Manager, “Strengthening Mobility and Promoting Regional Integration of Engineering Education in Africa (SPREE)” – €1,392,975 funded by the European Commission
2020 – 2021

APPLIED & SCIENTIFIC RESEARCH (SELECTED)

Co-PI, "Value chain analysis for enhancing the productivity of textile and garment industries in northern Ethiopia" – ETB 499,570, Mekelle University 2019 – 2021

Co-PI, "Sustainable Development of Textile and Garment Industries in Tigray, Ethiopia" – ETB 484,394, International Labour Organization 2016 – 2017

PI, "Design and development of a sesame and other seed oil extracting machine for sesame cultivators" – ETB 94,444, Mekelle University 2016 – 2017

PI, "Improving the productivity and competitiveness of Ethiopian leather and leather products industries" – ETB 105,000, Mekelle University 2015

Co-PI, "Improving energy efficiency of cookware and bake products in Ethiopia" – ETB 75,000, Mekelle University 2014

PI, "Identification and elimination of work-in-process wastes in Ethiopian textile and garment industries" – ETB 134,794, Mekelle University 2012 – 2013

INDUSTRIAL CONSULTANCY

Industrial Engineer, "Upgrading HR Job Grades, Career Paths, Job Titles, and Salary Scales for Mesfin Metals & Engineering PLC" – ETB 450,000, MESFIN Feb–May 2026

Industrial & Manufacturing Engineer, "Inspection and identification of detailed maintenance issues of heavy-duty machinery, Tigray Roads Authority" – ETB 1,903,875 Apr–Jul 2025

Industrial Engineer, "Comprehensive Feasibility Study for establishing a Galvanized Steel Factory (Mesfin Industrial Engineering)" – ETB 450,000 Feb–Jun 2025

Industrial Automation & Manufacturing Consultant, "Feasibility study for Olive Pharmaceutical manufacturing plant, specialized hospital, cardiac center, and medical supply wholesaling" – ETB 972,900, Olive Pharmaceutical Industries S.C. May–Sep 2024

Industrial Engineer, "Investment feasibility study for establishing Metal Casting Technology (Mesfin Industrial Engineering PLC)" – ETB 387,000 Nov 2023 – Jan 2024

Industrial Engineer, "Environmental Management Plan study services for Messebo Cement Factory" – ETB 1.2 million 2015

COMMUNITY SERVICES & TECHNOLOGY TRANSFER

TT/CS: Damage impact assessment for three METEC factories, organized by the Tigray Bureau of Industry (Co-PI); Mekelle, Ethiopia 2022

Coordinator, mega project proposal for job creation and income generation using small and medium manufacturing enterprises, Tigray Development Association; Mekelle, Ethiopia 2022

Coordinator, maintenance of high school and primary school desks and armchairs, Tigray Bureau of Education; Mekelle, Ethiopia 2022

TT: Design and development of a silk spinning machine for Tigray TVET workers; Mekelle, Ethiopia 2015

CS: Improving manufacturing flow using lean production tools for Maichew Particleboard Manufacturing Factory managers (PI); Maichew, Ethiopia 2014

CS: National policies and strategies, higher education mission, and Ethiopian renaissance for Mekelle University students; Ethiopia 2014

TT: Improving manufacturing flow using lean production techniques for textile and garment industries in Kombolcha and Addis Ababa, Ethiopia 2013

TT: Improving manufacturing flow using lean principles for MAA Garment and Textile managers; Mekelle, Ethiopia 2012

PUBLICATIONS

Peer-reviewed journal articles, conference papers, and a doctoral dissertation (selected):

1. Gidey, G., Mezgebe, T. T., & Abate, K. M. (2026). Taguchi design method: A parametric investigation and optimization of surface roughness of St 52-3 mild steel in a vertical knee-type milling operation. *Journal of Micro Manufacturing*, 9(2). <https://doi.org/10.1177/25165984261458547>
2. Haileselassie, B. M., & Mezgebe, T. T. (2026). Multicriteria decision-making for solving the problem of contractor selection: A case study in a road construction project. *International Journal of Multicriteria Decision Making*, 10(3). DOI: 10.1504/IJMCDM.2025.10075304 (in press).
3. Mezgebe, T. T., Gebreslassie, M. G., & Mittal, V. (2025). Localising digital manufacturing in Sub-Saharan Africa: challenges and strategic pathways for industrial competitiveness. *Discover Applied Sciences*. <https://doi.org/10.1007/s42452-025-07972-5>
4. Palanisamy, A. K., Jegatheeswaran, D., Ananth, C., Bhogayata, A., Juneja, B., Buribay, R., & Mezgebe, T. T. (2025). Prediction of early age compressive strength of concrete using machine learning. *Scientific Reports*. <https://doi.org/10.1038/s41598-025-29233-6>
5. Mezgebe, T. T., Tesfay, G., Amare, Z., & Alemayehu, R. (2024). Post-war recovery and growth eco-system for small and medium scale manufacturing enterprises in Tigray, Northern Ethiopia. *Branna Journal of Engineering and Technology*, 3(2).
6. Mezgebe, T. T., & Gebremariam, G. K. (2024). Lean production principle for improving productivity: Empirical case study in a garment industry in Ethiopia. *Jordan Journal of Mechanical and Industrial Engineering*, 18(1), 241–250. <https://doi.org/10.59038/jjmie/180119>
7. Sibhato, H., Gebreslassie, M. G., & Mezgebe, T. T. (2023). Contextualizing entrepreneurship for Africa's post-COVID-19 recovery and growth. *Scientific African (Elsevier)*, 22, e01946. <https://doi.org/10.1016/j.sciaf.2023.e01946>
8. Mezgebe, T. T., Gebreslassie, M. G., Sibhato, H., & Bahta, S. T. (2023). Intelligent manufacturing ecosystem: a post COVID-19 recovery and growth opportunity for manufacturing industry in sub-Saharan countries. *Scientific African (Elsevier)*, 19, e01547. <https://doi.org/10.1016/j.sciaf.2023.e01547>
9. Gebreslassie, M. G., Bahta, S. T., Mulugetta, Y., Mezgebe, T. T., & Sibhato, H. (2023). The need to localize energy technologies for Africa's post COVID-19 recovery and growth. *Scientific African (Elsevier)*, 19, e01488. <https://doi.org/10.1016/j.sciaf.2022.e01488>
10. Zimmermann, E., Mezgebe, T. T., El-Haouzi, H. B., Thomas, P., Pannequin, R., & Noyel, M. (2021). Multi-criteria decision-making method for scheduling problem based on smart batches and their quality prediction capability. *Computers in Industry (Elsevier)*, 133, 103549. <https://doi.org/10.1016/j.compind.2021.103549>
11. Mezgebe, T. T., & El-Haouzi, H. B. (2020). Recursive hybrid control architecture to deal with reactivity in the context of industry 4.0. 13ème Conférence Internationale de Modélisation, Optimisation et Simulation (MOSIM'20), Agadir, Morocco. <https://hal.science/hal-03083251>
12. Mezgebe, T. T. (2020). Human-inspired algorithms for designing new control system in the context of factory of the future (Doctoral dissertation, Université de Lorraine). NNT: 2020LORR0006. <https://hal.univ-lorraine.fr/tel-02510677>
13. Asgedom, H. B., Mezgebe, T. T., Beyen, K. T., & Mwasiagi, J. I. (2020). Investigation of industrial relations and working conditions of textile and garment factories in Northern Ethiopia. *Branna Journal of Engineering and Technology*, 1(2), 143–162.
14. Mezgebe, T. T., El-Haouzi, H. B., Demesure, G., & Thomas, A. (2019). Multi-agent systems negotiation to deal with dynamic scheduling in disturbed industrial context. *Journal of Intelligent Manufacturing (Springer)*, 31(6), 1367–1382. <https://doi.org/10.1007/s10845-019-01515-7>
15. Mezgebe, T. T., Demesure, G., El-Haouzi, H. B., Pannequin, R., & Thomas, A. (2019). CoMM: A consensus algorithm for multi-agent based manufacturing system to deal with perturbation. *International Journal of Advanced Manufacturing Technology (Springer)*, 105(9), 3911–3926. <https://doi.org/10.1007/s00170-019-03820-0>
16. Asgedom, B. H., Mezgebe, T. T., & Desta, A. (2019). Analyzing the causes of rejection and downgrading of semi-finished products in Ethiopian leather industries: situational analysis from the Sheba leather industry. *International Journal of Economic Research*, 7, 92–119.
17. Asgedom, H. B., Mwasiagi, J. I., & Mezgebe, T. T. (2019). Availability of labor for textile and apparel factories in Ethiopia: a case study of the Tigray region. *Annals of the University of Oradea, Fascicle of Textiles, Leatherwork*, 20(2).
18. Mezgebe, T. T., El-Haouzi, H. B., Demesure, G., & Thomas, A. (2018). A negotiation-based control approach for disturbed industrial context. *IFAC-PapersOnLine (Elsevier)*, 51(11), 1255–1260. <https://doi.org/10.1016/j.ifacol.2018.08.360>
19. Mezgebe, T. T., El-Haouzi, H. B., Demesure, G., Pannequin, R., & Thomas, A. (2018). A negotiation scenario using an agent-based modeling approach to deal with dynamic scheduling. In *Service Orientation in Holonic and Multi-Agent Manufacturing (Vol. 762, pp. 381–391)*. https://doi.org/10.1007/978-3-319-73751-5_29

20. Desta, A., Mezgebe, T. T., Berhe, H., & Gebresas, A. (2016). Kaizen as a strategic choice for enhancing the capabilities of manufacturing companies in Ethiopia. *International Journal of Business Strategy*, 16(1), 55–72.
21. Gebresas, A., Asmelash, H., Berhe, H., & Tesfay, T. (2015). Briquetting of charcoal from sesame stalk. *Journal of Energy* (Hindawi). <https://doi.org/10.1155/2015/757284>
22. Mezgebe, T. T., Asgedom, H. B., & Desta, A. (2013). Economic analysis of lean wastes: case studies of textile and garment industries in Ethiopia. *International Journal of Academic Research in Business and Social Sciences*, 3(8), 101. <https://doi.org/10.6007/IJARBS/v3-i8/123>

CONFERENCES, SYMPOSIUMS & WORKSHOPS

13th International Conference on Modeling, Optimisation and Simulation (MOSIM 2020); Agadir, Morocco	2020
STP du GDR MACS: Modélisation, Analyse et Conduite des Systèmes Dynamiques; Bordeaux, France	2019
Doctoral Workshop on Application of Artificial Intelligence in Manufacturing 4.0 (University of Lorraine & Tongji University); Nancy, France	2019
16th Symposium on Information Control Problems in Manufacturing (INCOM); Bergamo, Italy	2018
STP du GDR MACS: Modélisation, Analyse et Conduite des Systèmes Dynamiques; Lille, France	2018
7th International Workshop on Service Orientation in Holonic and Multi-Agent Manufacturing (SOHOMA'17); Nantes, France	2017
National Symposium on Science, Technology, and Innovation for National Development; Wollega University, Ethiopia	2014

PROFESSIONAL MEMBERSHIPS & COMMITTEES

Member, Ethiopian Industrial Engineers Association	2015 – Present
Member, French Doctoral Alumni	2020 – Present
Advisory Committee Member, 2nd International Conference on Engineering and Technology, Mekelle University, Ethiopia	2025
Task Force Member, implementation of ISO 9001:2008 Quality Management System, Mekelle University, Ethiopia	2015

SKILLS

Languages: Tigrigna – Mother tongue; English – Excellent (listening, reading, speaking, writing); Amharic – Excellent (listening, reading, speaking, writing); French – Good (listening, reading, speaking); Satisfactory (writing)

Software: SCADA, MATLAB, Python, Java, TeX/LaTeX, SolidWorks

Statistical tools: Minitab, MATLAB, ClickCharts, SPSS

REFERENCES

- Dr. Muluaem Gebregiorgis Gebreslassie – Associate Professor of Energy, Sheffield Hallam University, UK. M.Gebreslassie@shu.ac.uk
- Prof. Hind Bril El-Haouzi – Full Professor of Industrial Engineering, CRAN, University of Lorraine, France. hind.el-haouzi@univ-lorraine.fr
- Dr. Alula Gebresas – Associate Professor of Materials Science and Technology, EIT-M, Mekelle University, Ethiopia; Postdoctoral Researcher, Technical University of Clausthal, Germany. alula.gebresas@tu-clausthal.de